

SensoPack — Team Preparation & Q&A Handbook

Team Sentinel · IIT Kharagpur Platinum Jubilee Innovation Challenge · Round 2

How to use this handbook. Read the front pages once (they are shared by everyone). Then go to **your own person section** — each of your slides has three things together: **(1) the speech to say, (2) the detailed concept to understand, and (3) 3–4 questions** a judge may ask on that slide, with the answer. Also skim the other people's slides once, because two of our six members are not presenting — so any of the four of us must be able to answer for the whole project. Keep language simple. **Say only what is on the slides. Never claim what is not built yet.**

Big picture (everyone must know)

Problem: India is the world's largest shrimp exporter, but shrimp spoils very fast and cold chains often break. Normal packaging only *protects* — it cannot *tell* if the food is still fresh.

Our idea (SensoPack): one smart, biodegradable film that both **protects** the shrimp and **reports** its freshness (a colour change you can see, plus a sensor). A reusable NFC tag + cloud AI then predict shelf life and give a simple verdict — **Safe, Caution, or Unsafe**.

Why new: rivals do only one thing (protect OR report), often costly or with plastic waste. SensoPack combines **protection + sensing + biodegradable + low cost** in one film.

Honest stage: today we have a working **lab-scale bioplastic (biopolymer) film** (TRL 3–4). Chitosan from shrimp-shell waste is our **target** material; the sensor, the electronic tag and the AI are **designed** — our next steps.

Who presents what

Presenter	Slides	Topic
Naveen	1, 2, 3, 4	Intro · problem & market · why others fail & our solution · competitive edge
Jaspreet	5, 6	The science of spoilage · the sensing & manufacturing technology
Viraj	7, 8, 9	System architecture · impact · prototype & roadmap
Aishik	10, 11, 12	Business viability · implementation flow · close & Q&A

Team of 6; two not presenting (Prem Pastagia, Zalavadiya Dishil). If asked "who did what": *"We are a six-member team from different departments; four of us present today, two contributed to the work but could not attend."*

TRUE now vs. DESIGNED (memorise)

Built & shown today — say "we have"	Designed / next step — say "we will / our design"
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A transparent bioplastic (biopolymer) film — high optical clarity, clean edges, durable (TRL 3–4)	Moving to the target chitosan (shrimp-shell) + essential-oil formulation
The science base (biomarkers & thresholds — from literature)	Putting the colour indicator + MIP sensor inside the film
	The reusable NFC tag + low-power chip, and the cloud AI model
	Pilot testing on real refrigerated shrimp

Golden line if pushed: *"Our built prototype today is a transparent bioplastic film. The chitosan formulation, the sensor, the tag and the AI are our designed architecture and next milestones — and we are honest that we are early stage."*

Key numbers — cheat sheet

Fact	Number
India seafood exports (FY 2025–26)	US\$ 8.46 billion
Frozen shrimp share	US\$ 5.62 billion (66.52%)
Shrimp share of India's seafood earnings	~69%
India → USA shrimp (FY25)	US\$ 2.71 billion
Lost to spoilage per year	~₹15,000 crore
Global seafood lost before a plate	~35%
IoT tag cost (too high)	> \$2.50 / unit
Sensor must stay within	5–8% of package value
Our film cost (tiers)	under ₹5 up to ~₹120
Film thickness / tag reuse / markdown avoided / stage	<20 µm · 100–500 cycles · 20–25% · TRL 3–4

Source of market numbers, if asked: India's official marine-product export figures, FY 2025–26 (MPEDA). Keep it on your phone.

Key words — simple meanings

Term	Simple meaning
TVB-N	Smelly nitrogen gases (like ammonia) from protein breakdown. More = more spoiled. Limit ≈ 30–35 mg/100 g.
TMA	The "fishy smell" gas from bacteria. Limit ≈ 3–5 mg/100 g.

pH shift	Spoiling shrimp becomes less acidic (pH rises above 7.5). We can measure it.
Anthocyanin indicator	A natural colour that changes with pH. As gases raise the pH, it goes purple → green → brown. Freshness by eye.
MIP sensor	Molecularly Imprinted Polymer — an artificial "lock" that fits one molecule (TMA). Like a synthetic antibody.
CMOS ASIC (130 nm)	A tiny, cheap, very-low-power chip that reads the sensor and makes clean digital data.
NFC	Wireless chip read by tapping a phone. The read needs no battery — the phone powers it.
Chitosan	Natural film material from shrimp/crab shell waste; biodegradable, food-safe. Our target material — current lab film is a bioplastic.
Bioplastic / biopolymer	A biodegradable, plastic-like film from biological material. This is our current prototype.
MAP	Modified Atmosphere Packaging — changes the gas inside to slow spoilage. Protects but cannot report freshness.
TRL 3–4	Technology Readiness Level. 3–4 = working lab prototype / proof of concept. We are here.

Naveen — Slides 1, 2, 3, 4

Your job: open the talk, make the problem feel big and real, and show our idea is the first to solve it fully and cheaply.

SLIDE 1 — Title & Team

Naveen

SAY THIS

"Good morning. We are Team Sentinel. Our project is **SensoPack** — smart and green packaging for shrimp. I am Naveen. My team is Aishik, Jaspreet, Viraj, Prem and Dishil. Today we will show you the problem, our solution, the science, the impact, and the business."

UNDERSTAND THIS

- One-line of SensoPack: a biodegradable film that **protects** shrimp **and** shows if it is still fresh.
- TEVIX is our guiding idea: **T**echnology guided by **E**cology, driven by **I**ntelligence, creating **V**alue with **I**mpact.
- Keep it warm and short; set up the flow so judges know what is coming.

LIKELY QUESTIONS

Q: What is SensoPack in one line?

A: A smart, biodegradable film that protects shrimp and also shows if it is still fresh.

Q: You are six — why four presenting?

A: We are a six-member team from different departments. Four present today; two contributed but could not attend.

Q: What does "SensoPack" mean?

A: "Senso" for sensing freshness, "Pack" for packaging — packaging that senses.

SAY THIS

"India is the world's biggest shrimp exporter. In 2025 to 26, our seafood export was **8.46 billion dollars**. Frozen shrimp alone was **5.62 billion dollars** — about 66 percent. But every year, about **15,000 crore rupees** is lost to spoilage. The reason is simple. Shrimp spoils very fast. And cold storage is not always reliable. Normal packaging only protects the food. It cannot tell us if the food is still fresh."

UNDERSTAND THIS

- **Why shrimp spoils fast:** it is high in water and protein/nutrients — a perfect food for bacteria, so they multiply quickly.
- **Cold chain:** the chain of cold storage from catch to shop. One unnoticed temperature break causes spoilage; ~90% of our sector has no continuous cold chain.
- **The core gap:** packaging protects but is "blind" — it cannot report freshness to anyone.

LIKELY QUESTIONS**Q: Why is this a big problem for India specifically?**

A: India is the world's largest shrimp exporter — \$8.46 B seafood, \$5.62 B shrimp — yet ~₹15,000 crore is lost to spoilage every year.

Q: Why does shrimp spoil so fast?

A: It is high in moisture and protein, so bacteria grow very fast after death.

Q: Where do these numbers come from?

A: India's official marine-product export figures for FY 2025–26 (MPEDA).

Q: What exactly does "cold chain break" mean?

A: Any point where the shrimp gets warm — a delay, a power cut, bad transport — and no one records it. That hidden warming spoils the product.

SAY THIS

"So why is this problem not solved yet? Old solutions are costly or weak. Electronic tags cost more than **2.5 dollars** each — too costly for cheap packaging. Also, **90 percent** of our food sector has no proper cold chain. Our idea is different. First, our film both protects the food and shows its freshness. Second, we make the film from shrimp shell waste, so it is eco-friendly. Third, it works for small sellers and big exporters, both."

UNDERSTAND THIS

- **Cost barrier:** a freshness sensor must be only 5–8% of pack value; a \$2.50 electronic tag is far too costly for everyday packaging.
- **Our 3 ideas:** (1) dual-function film — protect + report in one; (2) circular design — from shrimp-shell waste + a reusable NFC tag, cutting plastic and e-waste; (3) tiered platform — cheap colour version for small vendors, up to AI + NFC for exporters.
- **Careful (material):** shrimp-shell waste → **chitosan** is our *target* design material. Our current lab film is a **bioplastic** — see Slide 9.

LIKELY QUESTIONS**Q: Why can't existing electronic tags solve this?**

A: They cost over \$2.50 per unit; the sensor must be only 5–8% of pack value, so they are too costly for mass packaging.

Q: What are your three key ideas?

A: Protect + report in one film; a circular design from shrimp-shell waste with a reusable tag; and a tiered platform from cheap colour to AI + NFC.

Q: Have you already made the film from shrimp waste (chitosan)?

A: Chitosan from shrimp waste is our target circular material. Our current lab film is a bioplastic; moving to chitosan is a development step.

SAY THIS

"This table shows why we are special. Other options do only one thing. Some extend shelf life but cannot report. Some report but need a scanner. IoT tags only check temperature and are costly. **SensoPack does all four things together** — it protects, it reports, it fits the pack, and it is biodegradable. And it starts from under **5 rupees**. Now Jaspreet will explain the science."

UNDERSTAND THIS

- The four tests in the table: **extends shelf life · reports freshness · packaging-integrated · biodegradable**.
- MAP protects but cannot report. Colour labels report but often need a scanner and don't preserve. IoT tags only track temperature, are costly, and not biodegradable.
- **Our novelty = the combination**. No single rival passes all four; SensoPack does, from under ₹5 (basic tier).

LIKELY QUESTIONS**Q: How is this different from freshness / colour labels?**

A: They only report (often need a scanner) or only protect. Ours does both in one biodegradable, low-cost film.

Q: What is your defensible novelty?

A: The combination — protect + sense + biodegradable + affordable in one product. No rival does all four.

Q: Is the under-₹5 cost real?

A: That is the basic colour tier; the full range is ₹5–120. It is a design target kept within 5–8% of pack value.

Jaspreet — Slides 5, 6

Your job: prove we understand the science of spoilage, and that our way to sense it is real and can be made at scale.

SLIDE 5 — Four-stage spoilage cascade

Jaspreet

SAY THIS

"Thank you, Naveen. Now the science. Shrimp does not spoil in one step. It spoils in **four stages**. First, the muscle becomes stiff. Second, its own enzymes break it down. Third, bacteria grow and make gases like TMA and ammonia. Fourth, the fat turns bad and black spots come. Each stage makes some chemicals we can measure. These chemicals are our sensing targets. This science is from a 2023 research paper."

UNDERSTAND THIS

- **The four stages:** 1) Rigor mortis (muscle stiffens); 2) Autolysis (its own enzymes break tissue into amino acids); 3) Microbial spoilage (bacteria like *Pseudomonas* make TMA, ammonia, biogenic amines); 4) Lipid peroxidation & melanosis (fats go rancid, black spots).
- **Biomarkers & thresholds:** TVB-N 30–35, TMA 3–5 (mg/100 g), pH above 7.5. These rise as it spoils, so measuring them = measuring freshness.
- Reference: **Das & Mishra, 2023** — keep this name ready.

LIKELY QUESTIONS

Q: What are the four stages?

A: Rigor mortis, autolysis, microbial spoilage, and lipid peroxidation & melanosis.

Q: What is a biomarker, and why measure it?

A: A chemical released as shrimp spoils (TMA, TVB-N, pH). They have known spoilage limits, so measuring them gives an objective freshness reading.

Q: What are the threshold numbers?

A: TVB-N 30–35 and TMA 3–5 mg per 100 g, pH above 7.5.

Q: Where is this science from?

A: Published research — Das & Mishra, 2023.

SAY THIS

"How do we detect these chemicals? We track three markers — **TMA**, **TVB-N**, and **pH**. For sensing, our design uses a special polymer sensor, a low-power chip, and **NFC**. NFC means you can read it with just a phone — no battery needed. The film is very thin, under **20 microns**. It can be made on normal packaging machines. No new factory is needed. Now Viraj will continue."

UNDERSTAND THIS

- **Detection:** we target TMA, TVB-N and pH — all validated freshness markers that rise together.
- **Sensing design:** a **MIP sensor** (artificial lock that binds TMA selectively), a **130 nm CMOS chip** (cleans the signal, ultra-low power), and **NFC** (phone-tap read, no battery for the read).
- **Manufacturing:** film under 20 microns, made by roll-to-roll / slot-die, works with existing heat-sealers — no new factory.
- Say "our sensing **design** uses..." — these parts are designed; the built item is the film (Viraj, Slide 9).

LIKELY QUESTIONS**Q: What is a MIP sensor, simply?**

A: A Molecularly Imprinted Polymer — an artificial lock shaped to fit one molecule like TMA. It binds it selectively, like a synthetic antibody.

Q: How can NFC work with no battery?

A: For the read, the phone's own signal powers the NFC tag. No battery is needed for the freshness read.

Q: Can this really be mass-produced?

A: By design, yes — a thin film made by roll-to-roll coating that fits existing heat-sealers, so no new factory. Holding quality on industrial rolls is a known scale-up step.

Q: Why these three markers?

A: TMA, TVB-N and pH are standard, validated seafood-freshness indicators that rise together as it spoils, giving a reliable picture.

Viraj — Slides 7, 8, 9

Your job: show the full system as one clear picture, show it matters (impact) and is real (prototype), and be the honest voice about our stage.

SLIDE 7 — System architecture (two-sheet)

Viraj

SAY THIS (SHORT)

"Thank you, Jaspreet. Our system has **two sheets**. Sheet one is the **film** — it protects the shrimp and also senses freshness, changing colour as it spoils. Sheet two is the **electronics** — a reusable NFC tag and cloud AI that read the data and predict the shelf life. Together they give a simple result: **Safe, Caution, or Unsafe**. One honest point — today we have built only the film. The tag and AI are our next steps."

UNDERSTAND THIS

- **Sheet 1 = Protection + Sensing (the film):** a disposable, food-safe biofilm that protects the shrimp and senses spoilage — a colour indicator (purple→green→brown) + a MIP sensor for TMA. Biodegradable.
- **Sheet 2 = Communication + Prediction (the electronics):** a reusable NFC tag reads the data and uploads to the cloud; the AI gives freshness status, remaining shelf life, and an export-quality check. Tag reusable 100–500 times.
- **Closed loop:** Protect → Sense → Communicate → Predict → verdict (Safe / Caution / Unsafe) before spoilage.
- **You carry the honesty line:** only the film is built; the tag and AI are designed / next steps.

LIKELY QUESTIONS

Q: Explain the two sheets.

A: Sheet 1 is the film — it protects the shrimp and senses freshness. Sheet 2 is the electronics — a reusable NFC tag plus cloud AI that read the data and predict shelf life.

Q: What have you actually built so far?

A: Only the film (a bioplastic film). The sensor, the tag and the AI are designed and are our next steps.

Q: How does the colour freshness indicator work?

A: A natural pigment (anthocyanin) changes colour with pH. As the shrimp spoils, gases raise the pH, so it shifts purple → green → brown — freshness by eye, no device.

SAY THIS

"What is the impact? Three things. One — food safety: it stops people eating spoiled shrimp and reduces bad preservatives like formalin. Two — money: good freshness proof saves **20 to 25 percent** loss for fishermen and sellers. Three — reach: it works for small village harbours and big export chains, both. So it helps India's exports."

UNDERSTAND THIS

- **Food safety:** real-time TMA/TVB-N tracking helps stop spoiled seafood being eaten and reduces the use of unsafe preservatives like formalin.
- **Economics:** proven freshness avoids the usual 20–25% distress markdown sellers take on doubtful stock — protecting margins for fishermen, exporters, retailers.
- **Reach:** the tiered model works from a small village harbour up to a global container chain.

LIKELY QUESTIONS**Q: How does it improve food safety?**

A: It tracks spoilage in real time, so unsafe shrimp is caught early, and there is less reason to use dangerous preservatives like formalin.

Q: What is the 20–25% figure?

A: The usual price cut (distress markdown) sellers take when freshness is doubtful. Proof of freshness avoids that cut.

SAY THIS

"Where are we now? We are at **TRL 3 to 4** — a working lab prototype. It is a **transparent biopolymer film**. Our results are good — high optical clarity, clean edges, and good strength. Our next steps are — add the NFC sensing nodes, build the AI shelf-life model, and do pilot testing with refrigerated shrimp in real conditions. Now Aishik will finish."

UNDERSTAND THIS

- **Built:** a transparent **biopolymer (bioplastic) film** — high optical clarity, clean edges, durable/resilient. This is TRL 3–4 (lab proof of concept).
- **Honest gap:** the sensor, the NFC tag and the AI are **not** built yet — they are designed. Chitosan (from shrimp waste) is the target material we will move to.
- **Known hurdles:** clarity on thin industrial rolls, surviving heat-sealing, moisture barrier in cold storage, electronics integration.
- **Next milestones:** embed the sensing nodes, build the AI model, run pilot trials with refrigerated shrimp.

LIKELY QUESTIONS**Q: What is TRL 3–4?**

A: A working lab prototype / proof of concept, on a 1-to-9 readiness scale. The film exists; full integration and field trials are ahead.

Q: Is the prototype chitosan or a bioplastic? (likely trap)

A: Chitosan from shrimp-shell waste is our target material shown in the design. Our current lab prototype is a transparent bioplastic film that shows the properties we need. Converting to chitosan is a development step.

Q: Do you have accuracy data or real-shrimp results?

A: We have material results (clarity, clean edges, strength). We do not yet have sensor-accuracy numbers or real-shrimp trials — that validation is our next planned step.

Aishik — Slides 10, 11, 12

Your job: show it is a workable business, explain the simple end-to-end flow, and close with calm confidence.

SLIDE 10 — Viability of the business

Aishik

SAY THIS

"Thank you, Viraj. How will we earn? Three ways. One — we sell the smart films to companies. Two — we give reusable NFC tags on lease to cold chains. Three — we give a paid AI software service for freshness. So it is one full system — cheap film, reusable electronics, and cloud AI. And for the buyer, just **one phone tap** shows if the tray is fresh."

UNDERSTAND THIS

- **Three revenue streams:** (1) film sales (B2B to processors/exporters); (2) reusable NFC tags leased to cold-chain operators (we keep ownership, they pay to use); (3) SaaS subscription to the AI freshness portal (recurring income).
- **Value proposition:** one ecosystem — cheap film + reusable electronics + cloud AI — low-cost, modular, works with existing packaging lines.
- **Cost logic:** basic colour film under ₹5; higher NFC/AI tiers up to ~₹120 — kept within 5–8% of pack value.

LIKELY QUESTIONS

Q: How exactly do you make money?

A: Selling the smart films, leasing reusable NFC tags to cold-chain operators, and a recurring SaaS subscription for the AI freshness portal.

Q: Who is your first customer?

A: Shrimp processors and exporters — they lose the most to spoilage and rejection, so they gain the most from proof of freshness.

Q: Is the under-₹5 cost realistic?

A: That is the basic colour tier; the full range is about ₹5–120. It is a design target within 5–8% of pack value; final costing comes with pilot manufacturing.

SAY THIS

"This is how it works, step by step. Shrimp is packed. The film protects it. The sensor detects spoilage. The NFC tag sends data to the cloud. The AI checks it. And it gives a simple result — **Safe, Caution, or Unsafe**. It helps shrimp exporters, retailers, and food-safety officers. It improves exports and lowers losses. Everything is built around one simple value — keep food fresh, and prove it."

UNDERSTAND THIS

- **The flow:** Shrimp packed → Film protects → Sensor detects spoilage → NFC tag → Cloud AI → verdict: Safe / Caution / Unsafe.
- **Customers:** shrimp processors & exporters, retail & cold-chain logistics, food-safety regulators.
- **Benefits:** better export competitiveness, lower operational losses, safer food.

LIKELY QUESTIONS**Q: Explain the flow simply.**

A: Packed → film protects → sensor detects → NFC sends to cloud → AI decides → Safe / Caution / Unsafe.

Q: Who are the customers?

A: Shrimp processors and exporters, retail and cold-chain logistics, and food-safety regulators.

Q: What if the sensor gives a wrong reading?

A: By design we use more than one signal — colour, TMA and pH — as cross-checks, and the AI gives a band (Safe/Caution/Unsafe), not one fragile number. Validating this is a next step.

SAY THIS

"To finish — SensoPack is not just packaging. It is a freshness-check system for the whole seafood supply chain. It protects, senses, and predicts. We have a working prototype and a clear plan ahead. Thank you. We are happy to take your questions."

UNDERSTAND THIS

- End on the one big idea: **protect + sense + predict**, in one affordable, sustainable product.
- Be calm; pause; invite questions. Do not add new claims at the end.
- If a hard question comes, any of the four can step in — use the golden rules below.

LIKELY QUESTIONS**Q: Sum it up in one line.**

A: SensoPack protects shrimp and proves its freshness in real time, affordably.

Q: What is your biggest strength?

A: One biodegradable, low-cost film that both protects and reports — no rival does all four things together.

Q: What is next after this?

A: Integrate the sensor into the film, build the AI model, and run pilot trials with refrigerated shrimp.

Cross-cutting questions (any presenter)

Q: Why shrimp and not all seafood/meat?

A: Shrimp is India's largest, highest-value seafood export and spoils fast — the most valuable place to start. The idea can extend later to fish and meat.

Q: Do you have a patent?

A: Formal IP is a next step. Our defensible novelty is the integration — an active protective film that also senses freshness, biodegradable and low-cost.

Q: Will the essential oils change the taste of the shrimp?

A: They are used in very small amounts inside the film. Keeping it food-safe with no off-flavour is a formulation goal we test for.

Q: (A hard question you cannot answer)

A: Be calm and honest: "That is a good question. We have not tested that yet — it is part of our next validation step. Our proven result today is the film." Never bluff.

Golden rules for answering judges

- **Answer in one clear sentence first**, then one line of support. Do not ramble.
- **Only use facts on the slides or in this handbook.** If it is not here, do not invent it.
- **Separate "built" from "designed".** Film (bioplastic) = built. Chitosan, sensor, tag, AI = designed / next step.
- **"Not yet" is a strong answer:** "We have not tested that yet — it is our next step."
- **Any of the four can answer for the whole project.** If one is unsure, another steps in calmly.
- **Speak slowly and simply.** Short sentences beat fancy words.

Never say these (not true yet / not on slides)

- Do **not** say the sensor, NFC tag, chip or AI are built or tested. Only the film is built.
- Do **not** call the current prototype "chitosan" — the built film is a **bioplastic (biopolymer)**; chitosan is our design target.
- Do **not** give any accuracy %, shelf-life-extension number, or pilot result — we have none yet.
- Do **not** say "patented", "certified", "approved", or "guaranteed".
- Do **not** invent customers, orders, funding, or awards. Do **not** exaggerate ("revolution", "millions of lives").

Prepared for Team Sentinel — SensoPack. Master your own slides (speech + concept + questions); skim the others once. Keep the cheat sheet, glossary and "built vs designed" table open while you rehearse. Speak slow, stay honest, finish calm.